P1) $V(r \leq R) = -\frac{\beta}{4} r^2 + V_0$

(a) $-\nabla V = \vec{E}$

$$-\frac{\partial}{\partial r}\left(-\frac{\beta}{4} r^2 + V_0\right) = \frac{\beta r}{2}\hat{r} = \vec{E}(r<R)$$

$$\vec{\nabla}\vec{E} = \frac{\rho}{\varepsilon_0}$$

$$\frac{1}{r^2}\cdot\frac{\partial}{\partial r}\left(r^2 \cdot \frac{\beta r}{2}\right) = \frac{1}{r^2}\cdot\frac{3\beta r^2}{2} = \frac{3\beta}{2} = \frac{\rho}{\varepsilon_0}$$

$$\Rightarrow \boxed{\rho = \frac{3\beta\varepsilon_0}{2}}$$

$$\boxed{Q_{enc} = \rho\cdot V = \frac{\cancel{3}\beta\varepsilon_0}{\cancel{2}}\cdot\frac{\overset{2}{\cancel{4}}}{\cancel{3}}\pi R^3 = 2\pi\beta\varepsilon_0 R^3}$$

(b) Por simetría esférica $\vec{E} = E(r)\hat{r}$, usando Gauss:

- $r<R$:

  Calculado anteriormente $\boxed{-\nabla V = E = \frac{\beta r}{2}\hat{r}}$

- $R<r<R_1$:

  $\oint \vec{E}d\vec{S} = \frac{Q_{enc}}{\varepsilon_0}$ $\qquad E(r)\cdot 4\pi r^2 = \frac{2\pi\beta\varepsilon_0 R^3}{\varepsilon_0}$

$$\boxed{E(R<r<R_1) = \frac{\beta R^3}{2r^2}\hat{r}}$$

$R_1 < r < R_1$: $\vec{E} = 0$ conductor

$r > R_2$: $E(r > R_2) = E(R < r < R_1)$

conductor eléctricamente neutro

$$\boxed{E(r>R_2) = \frac{\beta R^3}{2r^2}\hat{r}}$$

(c)

$$\cancel{V(\infty)}^{\,0} - V(R) = -\int_R^\infty$$

$$\left(-\frac{\beta}{4}R^2 + V_0\right) = \int_R^\infty E\,dr$$

$$-\frac{\beta}{4}R^2 + V_0 = \int_R^{R_1} \frac{\beta R^3}{2r^2}dr + \int_{R_2}^\infty \frac{\beta R^3}{2r^2}dr$$

$$-\frac{\beta}{4}R^2 + V_0 = \frac{\beta R^3}{2}\left(-\frac{1}{r}\right)\Big|_R^{R_1} + \frac{\beta R^3}{2}\left(-\frac{1}{r}\right)\Big|_{R_2}^\infty$$

$$-\frac{\beta}{4}R^2 + V_0 = \frac{\beta R^3}{2}\left(\frac{1}{R} - \frac{1}{R_1}\right) + \frac{\beta R^3}{2R_2}$$

$$V_0 = \beta\left(\frac{R^2}{4} + \frac{R^2(R_1-R)}{2R_1} + \frac{R^3}{2R_2}\right)$$

$$\boxed{\beta = \frac{V_0}{\left(\frac{R^2}{4} + \frac{R^2(R_1-R)}{2R_1} + \frac{R^3}{2R_2}\right)}}$$

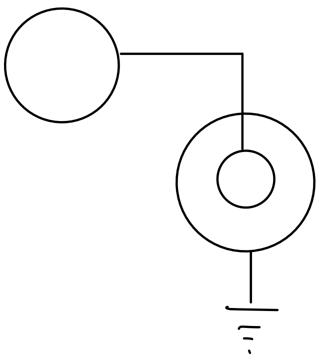
d) $R_1 < r < R_2$:

$$\oint \cancel{\vec{E}}^{\,0} \, d\vec{S} = \frac{Q_{enc}}{\varepsilon_0}$$

$$\left[\, 0 = Q_R + Q_{R1} \Rightarrow Q_{R1} = -2\pi\beta\varepsilon_0 R^3 \,\right]$$

$$\left[\, Q_{R2} = -Q_{R1} \Rightarrow Q_{R2} = 2\pi\beta\varepsilon_0 R^3 \,\right]$$

(e)



$$Q_R = 2\pi\beta\varepsilon_0 R^3 = Q_{fe} + Q_c \quad (1)$$

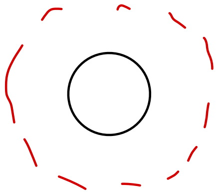
$$C = \frac{Q_c}{\Delta V}$$

C del condensador cilíndrico $\vec{E}(r_1 < r < r_2) = \dfrac{Q}{2\pi r L \varepsilon_0}\hat{r}$

$$\Delta V = \frac{Q_c}{2\pi L \varepsilon_0} \ln\left(\frac{r_2}{r_1}\right) \Rightarrow C = \frac{2\pi L \varepsilon_0}{\ln(r_2/r_1)}$$

$$\rightarrow \frac{2\pi L \varepsilon_0}{\ln(r_2/r_1)} = \frac{Q_c}{V_f} \Rightarrow Q_c = \frac{2\pi L \varepsilon_0 V_f}{\ln\left(\frac{r_2}{r_1}\right)} \quad (2)$$

carga final esfera



$$\vec{E} = \frac{Q_{fe}}{4\pi r^2 \varepsilon_0}$$

$$\underbrace{V(R^*)}_{V_{fe}} - \cancel{V(\infty)}^{0} = \int_{R^*}^{\infty} \frac{Q_{fe}}{4\pi r^2 \varepsilon_0} = \frac{Q_{fe}}{4\pi \varepsilon_0 R^*}$$

$$V_{fe} = \frac{Q_{fe}}{4\pi \varepsilon_0 R^*}$$

$$\Rightarrow Q_{fe} = 4\pi \varepsilon_0 R^* V_f \quad (3)$$

(2) y (3) en (1)

$$2\pi\beta\varepsilon_0 R^3 = \frac{2\pi L \varepsilon_0 V_f}{\ln\left(\frac{r_2}{r_1}\right)} + 4\pi\varepsilon_0 R^* V_f$$

$$\beta R^3 = V_f\left(\frac{L}{\ln(r_2/r_1)} + 2R^*\right)$$

$$V_f = \frac{\beta R^3}{\left(\frac{L}{\ln(r_2/r_1)} + 2R^*\right)}$$

$$\Rightarrow \boxed{Q_c = \frac{2\pi L \varepsilon_0}{\ln(r_2/r_1)} \cdot V_f = \frac{2\pi L \beta \varepsilon_0 R^3}{L + 2R^* \ln(r_2/r_1)}}$$